

COMPACT CY₃ COMPARISON LEDGER: EXTERNAL SCALAR THEOREMS, SUPPLIED COMPACT DATA, AND CENTRE-LIFT CONJECTURES

The formal-local mixed holomorphic–topological theory gives the Hamiltonian BF bulk on $\mathbb{R}_{\text{top}}^2 \times \widehat{\mathbb{C}}_{\text{hol}}^2$, the Dirac-brane trace $J(f) = \text{Tr} f(\phi_1, \phi_2)$, and the Capelli projective class ${}_{\text{W}}N[\bar{c}]$. A compact Calabi–Yau threefold carries a different but adjacent structure: a 3-Calabi–Yau category such as $\text{Perf}(X)$, virtual-cycle traces, the BCOV holomorphic anomaly, and Hodge-theoretic periods. A compact enumerative or automorphic comparison is a matched-conventions datum whose formal normal restriction may be compared with the displayed stalk. Its remaining coordinates are the compact virtual cycle, the Gromov–Witten/Pandharipande–Thomas comparison, the BCOV propagator, and a compact-target chain object whose protected trace has the automorphic scalar as its scalar invariant. This note is not used in the proof of the formal-Darboux trace-sector theorem; it is a compact-target comparison ledger.

I. MNOP FACTORISATION AND THE $\mathbb{E}_2$ -CENTRE OBSTRUCTION

On a smooth compact Calabi–Yau threefold X , three enumerative theories produce a priori distinct generating series: Donaldson–Thomas counts of ideal sheaves $Z_{\text{DT}}(\mathbf{q}, \mathbf{Q}) = \sum_{\beta, n} \text{DT}_{\beta, n}(X) \mathbf{q}^n \mathbf{Q}^\beta$ (Thomas 2000; Behrend–Fantechi 2008), Pandharipande–Thomas stable-pair counts $Z_{\text{PT}}(\mathbf{q}, \mathbf{Q}) = \sum_{\beta, n} \text{PT}_{\beta, n}(X) \mathbf{q}^n \mathbf{Q}^\beta$ (Pandharipande–Thomas 2009), and Gromov–Witten virtual-class series $Z_{\text{GW}}(u, \mathbf{Q}) = \exp \sum_{\beta, g} \text{GW}_{\beta, g}(X) u^{2g-2} \mathbf{Q}^\beta$ (Behrend–Fantechi 1997; Li–Tian 1998). The Maulik–Nekrasov–Okounkov–Pandharipande correspondence compares these series only after the reduced DT convention, the variable change $-q = e^{iu}$, and the multiple-cover normalisation are fixed. The scalar DT/PT factorisation is an external theorem in this form. The GW/PT comparison is a separate virtual-class comparison statement with its own hypotheses; neither statement is produced by the formal-local theorem.

Theorem 1.1 (External scalar DT/PT factorisation on the quintic). *Let $Q \subset \mathbb{P}^4$ be a smooth quintic threefold (Euler characteristic $\chi(Q) = -200$, Calabi–Yau by $c_1(Q) = 0$). The degree-zero DT generating function factors as the MacMahon power $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ raised to $\chi(Q)$. With $\mathbf{q}$ the Euler-characteristic variable and $\mathbf{Q}^\beta$ the Novikov monomial, the reduced DT series equals the PT series:*

$$Z_{\text{DT}}(Q; \mathbf{q}, \mathbf{Q}) = M(-\mathbf{q})^{\chi(Q)} Z_{\text{PT}}(Q; \mathbf{q}, \mathbf{Q}).$$

Proof. The factorisation $Z_{\text{DT}} = Z_{\text{DT},0} \cdot Z'_{\text{DT}}$ with $Z_{\text{DT},0} = M(-\mathbf{q})^{\chi(Q)}$ is Behrend–Fantechi’s Hilbert-scheme formula applied to the degree-zero locus. The reduced equality $Z'_{\text{DT}} = Z_{\text{PT}}$ is Bridgeland’s Hall-algebra theorem, with Toda’s wall-crossing route as auxiliary. Smoothness of Q and $\chi(Q) = -200$ fix the MacMahon exponent. $\square$

The PT/GW leg $Z_{\text{PT}}(Q; \mathfrak{q}, Q) = Z_{\text{GW}}(Q; u, Q)|_{-q=e^{iu}}$ is not a consequence of the scalar DT/PT theorem Theorem 1.1. For the quintic it is part of the GW/PT correspondence, constrained by Gopakumar–Vafa integrality and by the Huang–Klemm–Quackenbush bootstrap in the computed range.

The substitution $-q = e^{iu}$ is an analytic continuation datum between distinct generating functions. A chain-level lift must place the DT, PT, and GW theories in three dualisable modules over a factorisation-algebra image of $\text{Perf}(Q)$, each carrying its own trace; no scalar MNOP statement implies such a lift. The required compact comparison datum is

$$C_{\text{tar}}(X) = (\mathcal{F}_X, \text{Sp}_{\Sigma_2, C}^{\text{ch}}, \mathcal{A}_X, M_{\text{DT}}, M_{\text{PT}}, M_{\text{GW}}, \text{Tr}_{\text{DT}}, \text{Tr}_{\text{PT}}, \text{Tr}_{\text{GW}}, \mathfrak{C}_X^{\text{cent}}),$$

where $\mathcal{F}_X = \Phi_3^{\text{FA}}(\text{Perf}(X))$ denotes the required image of the 3-Calabi–Yau dg category of perfect complexes on X under a factorisation-algebra functor of degree 3; $\text{Sp}_{\Sigma_2, C}^{\text{ch}}$ is a chosen curve specialisation datum; $\mathcal{A}_X$ is the resulting chiral algebra after that specialisation; each $M_{\bullet}$ is a dualisable $\mathcal{A}_X$ -module attached to the corresponding theory; each $\text{Tr}_{\bullet}$ is a chain-level trace map recovering the scalar generating function; and $\mathfrak{C}_X^{\text{cent}}$ is the centre-comparison complex whose degree-zero Maurer–Cartan fibre classifies the central automorphisms compatible with these traces.

Conjecture 1.2 (MNOP centre-lift on the quintic). *Suppose $C_{\text{tar}}(Q)$ and the three chain-level trace maps are constructed in conventions matched to Theorem 1.1 and to the chosen GW/PT normalisation. With the curve specialisation and centre-comparison maps fixed, an MNOP lift is a point of the homotopy fibre*

$$\text{Lift}_{\text{MNOP}}(Q) := \text{hofib}\left(\text{Aut}_{E_2}(Z_{E_2}(\mathcal{A}_Q)) \longrightarrow \text{Aut}(\text{Tr}_{\text{DT}}, \text{Tr}_{\text{PT}}, \text{Tr}_{\text{GW}})\right)_{-q=e^{iu}}.$$

Thus the chain-level MNOP lift datum is defined by $C_{\text{tar}}(Q)$, the trace triple, and the displayed homotopy fibre. Non-emptiness requires the obstruction tower in $\mathfrak{C}_Q^{\text{cent}}$ to vanish. At a point of this fibre, acyclicity of the tangent complex in non-negative degrees gives uniqueness up to contractible choice.

The hypothesis decomposes into four mathematical data: a constructed factorisation-algebra functor Φ_3^{FA} on the 3-Calabi–Yau dg category, compatible with Hall and centre constructions; dualisable $\mathcal{A}_X$ -modules $M_{\text{DT}}, M_{\text{PT}}, M_{\text{GW}}$; chain-level trace maps whose protected characters realise the three scalar partition-function normalisation targets only after the corresponding trace complexes have been constructed; and centrality homotopies $\sigma_X: \text{Tr}_{\text{DT}} \Rightarrow \text{Tr}_{\text{PT}}$ and analogously to Tr_{GW} in the E_2 -centre. Existence is the lifting problem for this homotopy fibre, and rigidity is the acyclicity of its tangent complex. Orientation and cohomological-Hall-algebra (CoHA) prerequisites enter through Joyce–Upmeyer and Kinjo–Park–Safronov.

2. S^3 -FRAMING ON THE QUINTIC

The open topological theory with a Lagrangian brane $L \subset X$ requires a framing datum on L : a chosen trivialisation of the $SO(\text{rank } N_L)$ -torsor of orthonormal frames of the normal bundle, compatible with the brane-line BV complex. For a Lagrangian $L \cong S^3 \hookrightarrow Q$ in the

smooth quintic -- for instance the real locus of a real deformation (Sheridan 2016; Sheridan–Smith 2021) -- the topological framing obstruction lives in

$$\pi_0(\text{Map}(S^3, SO(3))) = \pi_3(SO(3)) = \mathbb{Z},$$

the Hirzebruch σ -invariant torsor. Two questions separate: whether the symplectic Lagrangian condition and an orientation of L trivialise $N_{L/Q}$ topologically, so that the framing torsor is $\mathbb{Z}$; and which integer in that torsor, paired with which BV/A_∞ chain homotopy on the brane complex, the open B-model requires.

Proposition 2.1 (Normal-bundle orientation and topological triviality). *Let $L \cong S^3 \subset Q$ be an oriented Lagrangian in the smooth quintic threefold. Then the oriented rank-three real bundle $N_{L/Q} \rightarrow S^3$ is topologically trivial. Its trivialisations form a $\pi_3(SO(3)) = \mathbb{Z}$ -torsor; no element of this torsor is determined by the Calabi–Yau volume form alone.*

Proof. The symplectic form identifies $N_{L/Q}$ with T^*L . The chosen orientation of L orients T^*L . Since S^3 is parallelisable, T^*S^3 is trivial. Every oriented rank-three real bundle on S^3 is also trivial by obstruction theory: the classifying map lies in $BSO(3)$ and $\pi_3(BSO(3)) = \pi_2(SO(3)) = 0$. $\square$

The topological construction gives the framing torsor, not a preferred framing. Two further data complete the open brane theory on (Q, L) : an integer framing class $[f] \in \pi_3(SO(3)) = \mathbb{Z}$ selecting a point in the trivialisation torsor, and a Costello-compatible BV/A_∞ chain homotopy h_f trivialising the obstruction class

$$\text{Ob}_{\text{frame}}(L, f) \in H^0(\Omega^\bullet(L) \otimes \text{End}_{BV})$$

to the equivariant lift of the brane propagator over the chosen framing. Framing on S^3 is a choice: each $[f]$ defines a different chain-level brane.

3. QUINTIC BPS GROWTH AND ABEL–JACOBI COMPARISON

If the genus-zero BPS counts n_N of degree- N rational curves on a compact Calabi–Yau threefold grow at the rate predicted by a vanishing period, the sequence $a_N = N^{-1} \log n_N$ has a finite limit. An Abel–Jacobi invariant is attached to a homologically trivial one-cycle, not to a curve class alone. Thus a compact CY-to-chiral theorem must choose homologically trivial cycles

$$\mathcal{Z}_N = C_{N\ell} - C_{N\ell}^\circ$$

and a Hodge norm on $J^2(X)$. The compact-target asymptotic conjecture is the assertion

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log |n_N(X, N\ell)| = \frac{1}{2\pi} \lim_{N \rightarrow \infty} \frac{1}{N} \|\mathbf{AJ}(\mathcal{Z}_N)\|_H,$$

where $n_N(X, N\ell)$ is the degree- N rational-curve count in class $N\ell \in H_2(X, \mathbb{Z})$, $\mathbf{AJ}(\mathcal{Z}_N) \in J^2(X)$ is the Abel–Jacobi period of the chosen homologically trivial difference, and the intermediate Jacobian $J^2(X) = H^3(X, \mathbb{C})/(F^2 H^3 + H^3(X, \mathbb{Z}))$ carries the Hodge metric. Take $X = Q$ the smooth quintic, $\beta = N\ell$ for $\ell \in H_2(Q, \mathbb{Z})$ the line class, and $n_N = Z_N(Q, N\ell)$

the genus-zero BPS invariant. The finite data through $N = 10$ (BCOV bootstrap for $N \leq 9$, Huang–Klemm–Quackenbush 2009 for $N = 10$) reads:

$$\begin{aligned} n_1 &= 2875 \\ n_2 &= 609\,250 \\ n_3 &= 317\,206\,375 \\ n_4 &= 242\,467\,530\,000 \\ n_5 &= 229\,305\,888\,887\,625 \\ n_6 &= 248\,249\,742\,118\,022\,000 \\ n_7 &= 295\,091\,050\,570\,845\,659\,250 \\ n_8 &= 375\,632\,160\,937\,476\,603\,550\,000 \\ n_9 &= 503\,840\,510\,416\,985\,243\,645\,106\,250 \\ n_{10} &= 704\,288\,164\,978\,454\,686\,113\,488\,249\,750. \end{aligned}$$

The sequence $a_N := N^{-1} \log n_N$ attains its minimum at $N = 3$:

$$\begin{aligned} a_1 &\approx 7.964, \quad a_2 \approx 6.660, \quad a_3 \approx 6.525, \quad a_4 \approx 6.554, \quad a_5 \approx 6.613, \\ a_6 &\approx 6.676, \quad a_7 \approx 6.733, \quad a_8 \approx 6.785, \quad a_9 \approx 6.832, \quad a_{10} \approx 6.873. \end{aligned}$$

The sequence decreases sharply through $N = 3$, then increases monotonically: $a_{N+1} - a_N > 0$ for $N \in \{3, \dots, 9\}$, with increments

$$\{+0.029, +0.060, +0.062, +0.058, +0.052, +0.046, +0.041\}.$$

Proposition 3.1 (Quintic BPS-growth finite calculation through $N = 10$). *For the displayed genus-zero quintic BPS table through $N = 10$, the sequence $a_N = N^{-1} \log n_N$ attains its minimum at $N = 3$ and, on the tail $N \in \{5, \dots, 10\}$, satisfies the three-parameter law*

$$a_N = L + \alpha \frac{\log N}{N} + \frac{\beta}{N}$$

with residual bounded by 5×10^{-6} and

$$L \approx 7.5925, \quad \alpha \approx -3.3204, \quad \beta \approx 0.4477.$$

The intercept L is numerically within three significant figures of the raw mirror-period ratio at the conifold point $z_c = 5^{-5}$ of the Candelas–de la Ossa–Green–Parkes (CDGP) mirror quintic:

$$|t(z_c)| = \left| \frac{\omega_1(z_c)}{\omega_0(z_c)} \right| \approx 7.5909,$$

where the fundamental period $\omega_0(z) = \sum_n \binom{5n}{n,n,n,n,n} z^n$ and the logarithmic period $\omega_1(z) = \omega_0(z) \log z + 5 \sum_n \binom{5n}{n,n,n,n,n} (H_{5n} - H_n) z^n$ (with H_k the k -th harmonic number) solve the quintic Picard–Fuchs operator $\mathcal{L}_{\text{PF}}$ of (1) below.

Proof. Linear least squares on the six equations $a_N = L + \alpha(\log N)/N + \beta/N$, $N \in \{5, \dots, 10\}$, gives the stated coefficients with maximum residual $4.62 \times 10^{-6} < 5 \times 10^{-6}$. The fundamental and logarithmic periods solve

$$(1) \quad \mathcal{L}_{\text{PF}} = \theta^4 - 5z(5\theta + 1)(5\theta + 2)(5\theta + 3)(5\theta + 4), \quad \theta = z\partial_z,$$

with $z = 5^{-5}\psi^{-5}$ the natural mirror coordinate. At $z_c = 5^{-5}$ the convergent power-series sum yields

$$\omega_0(z_c) \approx 1.0707258684, \quad \omega_1(z_c) \approx -8.1277650532,$$

giving $|t(z_c)| \approx 7.5908925831$. $\square$

Three numbers are visible in the finite table: the tail-fit intercept $L \approx 7.59$, the raw mirror-period ratio $|t(z_c)| \approx 7.5909$, and the attractor comparison value $5 \log 5 \approx 8.05$. They live in different normalisations. The fit shows only that the six displayed tail values are compatible with a period scale near $|t(z_c)|$; it does not identify the Abel–Jacobi normal function.

The Abel–Jacobi growth assertion

$$\lim_N N^{-1} \log n_N = (2\pi)^{-1} \lim_N N^{-1} \|\mathbf{AJ}(C_{N\ell} - C_{N\ell}^o)\|_H$$

is stated against the Hodge metric on $J^2(Q)$. The raw CDGP mirror ratio $t(z_c) = \omega_1(z_c)/\omega_0(z_c)$ is a period-coordinate ratio: ω_0, ω_1 sit in a CDGP basis of solutions of (1) before Hodge normalization, and the $2\pi i$ monodromy factor is not fixed. Bridging the two requires a normalisation datum

$$N_{\mathbf{AJ}}: (\omega_0, \omega_1, \omega_2, \omega_3) \mapsto (\varpi_0, \varpi_1, \varpi_2, \varpi_3)$$

matching the CDGP period basis to the Hodge filtration on H^3 and fixing the $2\pi i$ monodromy convention. The finite agreement $L \approx |t(z_c)|$ is a normalisation check relative to $N_{\mathbf{AJ}}$.

4. SERRE PAIRING AND THE COMPACT CALABI–YAU CATEGORICAL TRACE

The Serre functor and its pairing are the categorical Calabi–Yau datum that any compact CY₃ descent must preserve, before any module, tilting, or Hall-algebra structure is built. Van den Bergh’s noncommutative crepant resolution (NCCR) is an algebra $\text{End}_R(M)$ for a reflexive maximal Cohen–Macaulay R -module M , required to be homologically homogeneous over the singular Gorenstein base. On the smooth quintic there is no crepant-resolution datum; the compact descent begins with the Serre datum.

Proposition 4.1 (Serre pairing on smooth compact CY₃). *Let X be a smooth compact Calabi–Yau threefold. The Serre functor $S_X: D^b(X) \rightarrow D^b(X)$ is the shift $S_X = [3]$, and the Calabi–Yau pairing*

$$\text{RHom}(F, G) \otimes \text{RHom}(G, F) \longrightarrow \mathbb{C}[-3]$$

is non-degenerate on $\text{Perf}(X) = D^b(X)$.

Proof. Smoothness gives $\text{Perf}(X) = D^b(X)$. Serre duality is $S_X = \otimes K_X[\dim X]$; the Calabi–Yau condition $K_X = \mathcal{O}_X$ reduces this to $S_X = [3]$. The pairing is the Serre trace of $\text{RHom} \otimes \text{RHom} \rightarrow \text{RHom}(F, F)$ followed by the Serre trace $\text{RHom}(F, F) \rightarrow \mathbb{C}[-3]$. $\square$

The Serre datum $S_X = [3]$ is what a compact Calabi–Yau descent must preserve. The remaining global quasi-NCCR obstruction is the construction of a Serre-equivariant local algebraic or dg model $\mathcal{A}_X \rightarrow \text{End}_{D^b(X)}(\bigoplus_i F_i)$ generating $D^b(X)$, with finite-dimensionality of $\text{Ext}^*(F_i, F_j)$ and a descent datum identifying the local model with a global endomorphism sheaf compatibly with S_X . Such an object exists in the singular Gorenstein cases above; on the smooth quintic its existence is the Serre-equivariant compact-trace obstruction.

5. FOURIER–MUKAI PARTNERS FOR $K_3 \times E$

Kuznetsov homological projective duality (Kuznetsov 2007) starts from a smooth projective variety with a very ample line bundle inducing a non-trivial Lefschetz decomposition of D^b ; on any Calabi–Yau the Serre functor is a shift, and Bridgeland’s indecomposability theorem (Bridgeland 1999) constrains the only semiorthogonal decomposition to the trivial one. Both $D^b(K_3)$ and $D^b(E)$ are indecomposable, so the Lefschetz datum on $K_3 \times E$ is the trivial rectangular structure $\mathcal{A}_0 = D^b(K_3 \times E)$, and Kuznetsov Fano HPD lacks the required non-trivial Lefschetz datum. The derived-equivalence statement on $K_3 \times E$ takes a different shape: Fourier–Mukai partner construction governed by Mukai-vector divisibility.

Theorem 5.1 (Fourier–Mukai equivalence in the isotropic $K_3 \times E$ case). *Let S be a projective K_3 surface with primitive polarisation H_S , and let E be a principally polarised elliptic curve. Fix a Mukai vector $v = (r, \ell, s) \in \tilde{H}(S, \mathbb{Z})$ with $\ell \in \text{NS}(S)$, Mukai pairing $\langle v, v \rangle = 0$, and H_S chosen v -generic. Assume the fine-moduli condition*

$$(\star) \quad \text{div}(v) := \gcd\{\langle v, w \rangle : w \in \tilde{H}(S, \mathbb{Z})\} = 1.$$

If $\text{NS}(S) = \mathbb{Z}H_S$ and $\ell = mH_S$, this condition is $\gcd(r, mH_S^2, s) = 1$. Let $\mathcal{M} := \mathcal{M}_{H_S}(v)$ be the moduli space of H_S -stable sheaves with Mukai vector v . Then:

- (i) $\mathcal{M}$ is a smooth projective K_3 surface (Mukai 1987; O’Grady 1997);
- (ii) there is an equivalence of $\mathbb{C}$ -linear triangulated categories

$$\Phi_{\mathcal{U}} : D^b(S) \xrightarrow{\sim} D^b(\mathcal{M}),$$

given by Fourier–Mukai transform along the universal sheaf $\mathcal{U}$ on $S \times \mathcal{M}$ (Bridgeland–Maciocia 2001, Complex surfaces with equivalent derived categories, Thm. 1.1);

- (iii) *combined with the Poincaré–Mukai autoequivalence $\text{FM}_{\mathcal{P}} : D^b(E) \xrightarrow{\sim} D^b(\hat{E}) = D^b(E)$ (Mukai 1981), the product transform*

$$\Phi_{\mathcal{U}} \boxtimes \text{FM}_{\mathcal{P}} : D^b(S \times E) \xrightarrow{\sim} D^b(\mathcal{M} \times E)$$

is an equivalence of CY₃ categories preserving the Serre structure $S_X = [3]$. When $\mathcal{M} \simeq S$ this is an untwisted Fourier–Mukai self-equivalence; in general it is a derived equivalence to the partner $\mathcal{M} \times E$.

Proof. (1) Under $(\star)$ and v -genericity of H_S , every H_S -semistable sheaf with Mukai vector v is stable (zero strict semistables occur); the moduli $\mathcal{M}$ is then a smooth projective holomorphic symplectic variety of dimension $\langle v, v \rangle + 2 = 2$. Hence $\mathcal{M}$ is a K_3 surface. For positive Mukai square the moduli space is higher-dimensional hyperkähler, and the product with E is no longer a threefold.

(2) The universal sheaf $\mathcal{U}$ exists on $S \times \mathcal{M}$ under $(\star)$: primitive Mukai divisibility kills the Brauer obstruction to a universal family (Căldăraru 2000, Thm. 1.3.2). The Fourier–Mukai kernel $\mathcal{U} \in D^b(S \times \mathcal{M})$ satisfies the Bridgeland–Maciocia orthogonality criterion (Bridgeland–Maciocia 2001, §1) and induces an equivalence. Under a nonzero Brauer obstruction to $(\star)$, $\mathcal{U}$ exists as a $p_M^* \alpha$ -twisted sheaf for a Brauer class $\alpha \in \text{Br}(\mathcal{M})$, yielding instead $D^b(S) \simeq D^b(\mathcal{M}, \alpha)$ (Huybrechts–Stellari 2005).

(3) On E with principal polarisation, $\hat{E} \cong E$ canonically; the Poincaré bundle $\mathcal{P} \in D^b(E \times E)$ is the kernel of $\text{FM}_{\mathcal{P}}$ (Mukai 1981, Thm. 2.2). Tensoring kernels, $\mathcal{K} := \mathcal{U} \boxtimes \mathcal{P} \in D^b((S \times E) \times (M \times E))$ defines a Fourier–Mukai transform whose composition with its adjoint is the identity on each factor; the external-tensor convolution $(\Phi_1 \boxtimes \Phi_2) \circ (\Psi_1 \boxtimes \Psi_2) = (\Phi_1 \circ \Psi_1) \boxtimes (\Phi_2 \circ \Psi_2)$ for FM kernels (Huybrechts, *Fourier–Mukai transforms in algebraic geometry*, Prop. 5.10) lifts the factor identities to the product. The Serre functor is preserved by any exact equivalence of triangulated categories (uniqueness of the Serre functor up to natural isomorphism); since both $\Phi_{\mathcal{U}}$ and $\text{FM}_{\mathcal{P}}$ are equivalences between CY categories of dimensions 2 and 1 respectively, the product Serre $S_{S \times E} = S_S \boxtimes S_E = [2] \boxtimes [1] = [3]$ survives. $\square$

Proposition 5.2 (Brauer-twisted dichotomy in the isotropic case). *Keep the isotropic hypothesis $\langle v, v \rangle = 0$ and drop $(\star)$ in Theorem 5.1. Then there exists a Brauer class $\alpha \in \text{Br}(M)$, whose order divides $\text{div}(v)$, such that*

$$D^b(S \times E) \simeq D^b((M, \alpha) \times E),$$

where the right-hand side denotes the derived category of $p_M^* \alpha$ -twisted sheaves. The partner is $(M, \alpha) \times E$ with twisted K_3 factor; an untwisted kernel exists only under the fine-moduli hypothesis.

Proof. Căldăraru 2000 Thm. 1.3.2 constructs the Brauer obstruction to the universal sheaf from the non-fine Mukai vector; its order is controlled by $\text{div}(v)$. Huybrechts–Stellari 2005 (*Equivalences of twisted K_3 surfaces*) then identifies the derived equivalence as a twisted Fourier–Mukai. The elliptic factor contributes an untwisted $\text{FM}_{\mathcal{P}}$. $\square$ $\square$

The Kuznetsov–Perry 2019 categorical-join / rectangular-HPD machinery requires a non-trivial Lefschetz decomposition; here $D^b(K_3)$ and $D^b(E)$ are indecomposable, so the Fourier–Mukai statement remains the relevant equivalence.

Remark 5.3 (Distinct κ values on $K_3 \times E$ and its anchors). The fine-moduli condition $(\star)$ is the primitive Mukai-vector hypothesis controlling the untwisted Fourier–Mukai kernel. Four κ values attached to $K_3 \times E$ and its comparison anchors are pairwise distinct:

- $\kappa_{\text{cat}}(K_3 \times E) = \chi(\mathcal{O}_{K_3 \times E}) = 0$, the total-space categorical Euler invariant by Künneth.
- $\kappa_{\text{fibre}}(K_3) = \chi(\mathcal{O}_{K_3}) = 2$, the fibre Euler datum.
- $\kappa_{\text{Mukai}}(K_3) = \text{rk } \tilde{H}^*(K_3) = 24$, the rank of the extended Mukai lattice $\tilde{H}^*(K_3, \mathbb{Z}) = H^0 \oplus H^2 \oplus H^4$.
- $\kappa_{\text{ch}}^{\text{Heis}} = 3$, while the Borcherds square-root branch $\Delta_5 = \Phi_{10}^{1/2}$ has weight 5 and the Igusa cusp form Φ_{10} has weight 10.

The Fourier–Mukai equivalence of Theorem 5.1 preserves the categorical invariant; it does not identify these four comparison invariants.

6. NON-PERTURBATIVE BCOV: GOPAKUMAR–VAFA CLOSURE

Physics large-order arguments and one-parameter computations predict conifold factorial growth for compact BCOV amplitudes,

$$F_g \sim (2g - 3)! / A^{2g-2},$$

with $A = \mathcal{N}_{\text{per}}(Z_{\gamma_c})$ the normalized Borel-plane image of the vanishing-cycle central charge $Z_{\gamma_c} = \int_{\gamma_c} \Omega$ at the nearest conifold singularity (Shenker; Mariño et al.). A theorem requires a chosen BCOV solution, boundary data, Borel plane, and Stokes sector. Non-perturbative BCOV is a compact-target resurgent problem with its own Stokes data and compact period normalization. Two structures remain: the Gopakumar–Vafa BPS resummation, and a global resurgent Stokes sheaf.

Costello–Li 2012 (arXiv:1201.4501) construct the classical BCOV field theory and the perturbative BV quantisation problem, with elliptic-curve uniqueness and Virasoro results within the stated range. Write $\{F_g^{\text{BCOV}}(X, t)\}_{g \geq 0}$ for a solution of the BCOV holomorphic anomaly equation with chosen large-radius and conifold-gap boundary data; the perturbative generating function is

$$F^{\text{pert}}(\lambda, t) = \sum_{g \geq 0} \lambda^{2g-2} F_g(X, t), \quad \lambda = g_s.$$

The Gopakumar–Vafa expansion (Gopakumar–Vafa 1998) proposes the non-perturbative form

$$(2) \quad F^{\text{GV}}(\lambda, t) = \sum_{\beta > 0} \sum_{g \geq 0} n_{\beta}^g \sum_{m \geq 1} \frac{1}{m} (2 \sin(m\lambda/2))^{2g-2} e^{-m\beta \cdot t},$$

where $n_{\beta}^g \in \mathbb{Z}$ are integer BPS invariants of the class β . Each fixed- β summand resums infinitely many perturbative genera and produces Schwinger poles at $\lambda = 2\pi k/m$ whose residues are pair-production amplitudes in the dual gauge description.

Conjecture 6.1 (GV non-perturbative closure on the quintic). *Let $Q \subset \mathbb{P}^4$ be the smooth quintic, $F_g^{\text{BCOV}}(Q, t)$ the genus- g solution of the BCOV holomorphic anomaly recursion with the CDGP 1991 large-radius normalization and conifold-gap boundary condition, and F^{GV} the Gopakumar–Vafa resummation (2) with the locally mirrored explicit quintic data of Huang–Klemm–Quackenbush 2009 through genus 20, together with their boundary-determination assertion through genus 51. In the Kähler cone $\Re t > t_c$, where t_c is the image of the conifold locus $\psi = 1$ on the complex-structure moduli space of the mirror quintic $\tilde{Q}$:*

(i) *For every $0 \leq g \leq 20$ after the unstable $g = 0, 1$ constant-map conventions are fixed,*

$$F_g^{\text{BCOV}}(Q, t) = [\lambda^{2g-2}] F^{\text{GV}}(\lambda, t)$$

as formal power series in $e^{-\beta \cdot t}$ with rational coefficients. Integrality is the assertion that the inverse multiple-cover transform gives integers $n_{\beta}^g(Q)$.

(ii) *The analytic convergence obstruction ob_{conv} for the genus-0 BPS series $\sum_{\beta} n_{\beta}^0 e^{-\beta t}$ vanishes in the Hodge-normalised Kähler coordinate, giving a nonempty convergence domain and a coefficient-growth $\limsup$ bounded by the nearest singularity in that coordinate.*

(iii) *There is a resurgent comparison whose perturbative sector is F^{pert} , whose compact scalar target datum, when included, has logarithm F^{GV} , and whose conifold Stokes wall is compatible with the Candelas–de la Ossa–Green–Parkes 1991 mirror-period monodromy. In that comparison the instanton action is $A_{\text{BCOV}} = \mathcal{N}_{\text{per}}(Z_{\gamma_c})$, where $Z_{\gamma_c} = \int_{\gamma_c} \Omega_{\tilde{Q}}$; the convention $A_{\text{BCOV}} = 2\pi i Z_{\gamma_c}$ is one choice of $\mathcal{N}_{\text{per}}$. The Stokes constant S_c is part of the comparison datum.*

Clause (i) does not follow from Theorem 1.1. It requires the CDGP prepotential, the HKQ genus-20 explicit data and boundary-determination theorem, the GV multiple-cover transform, and the GW/PT comparison as independent data. Clause (ii) is a convergence problem for the genus-zero BPS series: Castelnuovo-type genus bounds control arithmetic genus, while the rational-curve numbers n_β° govern the GV series. Finite-genus and finite-degree data alone leave the coefficient growth of $\sum_\beta n_\beta^\circ e^{-\beta t}$ undetermined; the analytic input is a coefficient-growth theorem in the Hodge-normalised Kähler coordinate. Clause (iii) is the local conifold Stokes match: the leading factorial growth $F_g^{\text{pert}}(Q, t) \sim (2g-3)!/\mathcal{N}_{\text{per}}(Z_{\gamma_c})^{2g-2}$ along the Borel direction fixed by the vanishing cycle γ_c identifies the B-model Borel singularity with the vanishing-cycle hypermultiplet of mass $|Z_{\gamma_c}|$. A comparison with the sine poles of (2) is an additional resurgent mirror theorem; it does not follow from the scalar GV expansion.

Conjecture 6.2 (Conifold comparison on the quintic). *On the quintic $Q \subset \mathbb{P}^4$, there exists a resurgent mirror comparison with a chosen Borel coordinate, flat coordinate, holomorphic-volume normalisation, and central-charge convention in which*

$$A_{k,m} = \mathcal{N}_{\text{Sch}}(p_{k,m}), \quad A_c = \mathcal{N}_{\text{per}}(Z_{\gamma_c}).$$

Equality $A_{k,m} = A_c$ is asserted only for the pole indexed by the vanishing-cycle BPS charge selected by the comparison. Here A_c is the Shenker–Mariño instanton action of the perturbative BCOV series, read off as $|F_g|/|F_{g+1}| \sim |A_c|^2/(2g-2)(2g-1)$; $p_{k,m} = 2\pi k/m$ is the raw Gopakumar–Vafa λ -plane pole at multiplicity m ; $Z_{\gamma_c} = \int_{\gamma_c} \Omega_{\tilde{Q}}$ is the central charge of the vanishing cycle at the conifold locus $\psi = 1$ on the mirror quintic; and $\mathcal{N}_{\text{Sch}}, \mathcal{N}_{\text{per}}$ are the comparison maps to the BCOV Borel plane. The same comparison must compute the conifold Stokes constant and identify the protected scalar A -model BPS datum, if such a datum exists.

The proposed equality is not an equality of raw symbols. It is the large-order conifold statement after the normalisations $\mathcal{N}_{\text{Sch}}$ and $\mathcal{N}_{\text{per}}$ have been constructed. The comparison from GV sine-pole singularities to the BCOV Borel plane and the conifold Stokes constant are part of the same datum.

Conjecture 6.1(iii) specifies the Borel sum along the physical ray and the matching problem at the single conifold Stokes wall. Full non-perturbative completion requires matching at every Stokes wall of the quintic moduli space: large complex structure (where (2) is the target definition), the conifold (where Strominger’s hypermultiplet pairs with the vanishing cycle), and Landau–Ginzburg / Gepner (where the HKQ bootstrap gives finite-genus data). The required datum is a global resurgent sheaf $\mathcal{R}^{\text{np}}$ on the moduli space together with a section $F^{\text{np}}(\lambda, t)$ whose Stokes jumps match the BCOV monodromy representation on $H^3(\tilde{Q}, \mathbb{Z})$. Construction of $\mathcal{R}^{\text{np}}$ is the non-perturbative BCOV obstruction, both globally on the quintic moduli space and in any compact CY₃ generalisation.

Conjecture 6.3 (Mirror-period recovery from non-perturbative BCOV). *Let an integral symplectic period basis, monodromy normalisation, and $2\pi i$ -convention be fixed. Let $\varpi_0(z), \varpi_1(z), \varpi_2(z), \varpi_3(z)$ be the Hodge-normalised CDGP 1991 fundamental periods of the mirror quintic $\tilde{Q}$, solutions of the Picard–Fuchs operator $\mathcal{L}_{\text{PF}}$ of (1), with the mirror coordinate*

$z = 5^{-5}\psi^{-5}$ and conventions matched to those of Proposition 3.1. The genus-zero prepotential extracted from $F^{\text{GV}}(\lambda, t) = \lambda^{-2}\mathcal{F}_0(t) + O(\lambda^0)$ with mirror map $t = \varpi_1/\varpi_0$ satisfies

$$\mathcal{F}_0(t) = \frac{1}{\varpi_0^2}(\varpi_0\varpi_3 - \varpi_1\varpi_2) + (\text{classical polynomial}),$$

with instanton corrections encoded exactly in $\{n_\beta^0\}_\beta$. The first three BPS values $n_1^0 = 2875$, $n_2^0 = 609\,250$, $n_3^0 = 317\,206\,375$ come from the BCOV table; the CDGP period/prepotential row and the Schubert–Katz–Ellingsrud–Strømme historical mirrors enter as numerical data.

Under the Hodge-normalisation datum $N_{\mathbf{AJ}}$, the comparison asks for a normal function with chosen cycles $\mathcal{Z}_N$, regulator map, Hodge metric, and $2\pi i$ -convention. The number L in Proposition 3.1 is a finite tail-fit intercept. An Abel–Jacobi growth limit requires the normal function and Hodge norm; only then can one compare it with $|\varpi_1(z_c)/\varpi_0(z_c)|$. Finite evidence comes from the HKQ genus-20 explicit data; higher-genus extension requires the resurgent completion of Conjecture 6.1(iii).